

Sajid Mostafiz Noor

+880-1815-458659 — sajidmostafiznoor@gmail.com — Portfolio — LinkedIn

Education

Bangladesh University of Engineering and Technology (BUET)
B.Sc. in Computer Science and Engineering (CSE)
CGPA: 3.70 / 4.00

Dhaka, Bangladesh
2022 – 2026

Research Experience

Next-Generation Computing Lab, BUET CSE

2024 – Present

- Collected data using wearable sensors, performed thematic analysis, and surveys for ongoing lab projects.
- Analyzed and visualized data on the heat impact of rickshaw pullers including data preprocessing and result interpretation.

Undergraduate Thesis: Aggressive Driving Detection Using Machine Learning

2025 – Present

Department of Computer Science and Engineering, BUET

- Implemented real-time detection of aggressive driving using machine learning frameworks and used feature selection algorithms to reduce input dimensionality while maintaining high model accuracy.
- Built and evaluated ML models (Decision Tree, Random Forest, SVM) for efficient, real-time inference, including strategies to handle missing or incomplete sensor features.

Publications

Real Time Machine Learning Based Detection of Aggressive Driving using Feature Optimization

2025 – 2026

Ibnul Islam, MD Sajid Mostafiz Noor, Ashikur Rahman, Raqeebir Rab, Abderrahmane Leshob

International Conference on Intelligent Multimedia, Networking, and Security (doi: 10.1109/IMNS67862.2026.11655308)

- Developed a real-time aggressive driving detection system using optimized sensor features.
- Used three feature selection algorithms (Feature Inclusion, Feature Exclusion, and Feature Inclusion with Group Sequencing Algorithms) to reduce input dimensionality.
- Achieved 98.54% accuracy with 66% reduction in the number of features, along with a feature-sparsity aware inference mechanism for reliable prediction under missing sensor data to improve efficiency and deployability.

Forecasting Occupational Survivability of Rickshaw Pullers Under a Changing Climate

2023 – 2025

Rahaman, M., Hasana, M., Rahman, S. S., Noor, M. S. M., Abedin, R. R., Tahmid, M. T., Parris, D. W., Choudhury, T., Al Islam, A. B. M., & Rahman, T

ACM Interactive, Mobile, Wearable and Ubiquitous Technologies (doi: 10.1145/3770712)

- Analyzed physiological stress on rickshaw-pullers under extreme heat using wearable sensor data.
- Applied Bayesian regression to model occupational heat-risk and survivability and integrated CMIP6 climate projections to forecast long-term climate impact.
- Conducted data preprocessing, statistical modeling, result validation; performed thematic analysis using qualitative data collected from structured interviews with rickshaw pullers.

From Wearable Data to Risk Literacy: Visualizing Physiological Heat Stress for Rickshaw Pullers in Dhaka

2025–2026

Accepted in Proceedings of the 2026 ACM International Symposium on Wearable Computers (ISWC)

- Designed simple, threshold-based visualizations to communicate heat-related physiological risk from wearable sensor data to low-literacy informal workers.
- Conducted a mixed-method study with 11 rickshaw pullers in Dhaka, using a pre and post assessment to capture how their perceived heat risk shifted after the visualizations.
- Performed thematic analysis of interviews and used those insights to suggest practical heat-safety interventions.

Projects

Bangla Biology Textbook Retrieval Benchmark for Educational QA (GitHub)

January 2026 – May 2026

- Built a Bangla textbook retrieval benchmark for educational question answering using 3 HSC Biology chapters with 166 textbook nodes and 1,320 question-answer pairs.
- Trained and evaluated BM25, Multilingual-E5, and hybrid retrieval models using PyTorch and Hugging Face Transformers.
- Compared different chunking methods, hard-negative mining, and fusion techniques through controlled experiments, achieving the best result with hybrid MinMax fusion, reaching 0.746 MRR and 0.665 Recall@1.
- Created a reproducible training and evaluation pipeline for future Bangla NLP and educational AI research.

BUET Fantasy Premier League (Live)

July 2025 – September 2025

- Built a fantasy sports platform modeled after the Premier League FPL system for managing fantasy teams based on real players.
- Implemented backend services, fantasy point calculation algorithms, and leader-board synchronization using RESTful APIs.
- Enabled users to manage teams, make team changes, make transfers, view player statistics, fixtures, and match results.
- Deployed the platform on Vercel with Github CI/CD actions with optimized performance for better user experience.
- Supports a growing user base of 100+ registered users, including 82 active users involved in fantasy management.

ExamCraft – AI-driven Exam Preparation Platform (GitHub)

February 2025 – August 2025

- Developed an AI-powered platform for quizzes, flashcards, and mock exams for personalized learning.
- Implemented AI-based question generation and spaced-repetition for adaptive study planning.
- Added spaced-repetition learning and performance analytics for custom recommendations and progress tracking.
- Automated build and deployment processes with GitHub Actions and deployed the containerized application on a virtual machine.

C++ Compiler (GitHub)

January 2024 - March 2024

- Built a compiler from scratch in C++ using Flex (lexical analysis) and Bison (parsing).
- Designed grammar rules, symbol table management, and semantic analysis with proper error detection and reporting.
- Generated intermediate three-address code as a machine-independent representation for execution and optimization.

Grades Information System (GitHub)

June 2023 – September 2023

- Designed a role-based academic management system with modular student, teacher, and admin services ensuring controlled data access.
- Engineered core backend logic for course enrollment with prerequisite validation, grade management, and result processing.
- Created an admin dashboard with CRUD operations for managing students, teachers, courses, notices, and exam schedules, and implemented database logging to ensure data consistency and track system changes.

Awards & Scholarships

- **Dean's List Scholarship**, BUET – January 2022 (Level-1, Term-1) ; July 2021 (Level-1, Term-2)
- **Board Scholarships**, Secondary School Certificate (SSC) and Higher Secondary Certificate (HSC)

Skills

Programming Languages: Python, Java, C, C++, JavaScript, TypeScript**AI & Machine Learning:** Machine Learning, Deep Learning, TensorFlow, PyTorch, Scikit-learn**Frontend, Backend & APIs:** React, Next.js, Tailwind CSS, Node.js, NestJS, FastAPI, REST APIs**Databases & DevOps:** MySQL, MongoDB, Docker, Git, GitHub Actions, Vercel